# Additive Effects of Fixed-Length Sequences in Promoting DNA Demethylation

# DNA脱メチル化の促進における固定長配列の加法的効果

付和文抄訳

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# Abstract

DNA methylation is essential for gene regulation, but predicting its inheritance remains a challenge. While the deep learning model CMIC showed high accuracy, its black box nature limits biological understanding (Maruyama 2022). This study aims to use interpretable models such as Random Forest or Logistic Regression to identify DNA motifs that control methylation inheritance. We analyzed 673 mouse genomic regions (396 maintained, 277 demethylated) using k-mer method ($k = 3$ to $8$). The Logistic Regression model at $k = 6$ achieved an accuracy of 86.78% and an AUC of 0.930. The results lead to two major conclusions. First, the success of the linear model suggests that the additive effects of DNA motifs are dominant in this process. Second, factor analysis revealed that demethylation pressure, which is driven by CpG count and specific demethylation motifs exerts a more dominant influence than maintenance signals. These findings demonstrate that interpretable models can successfully uncover key biological mechanisms that govern DNA methylation inheritance.

# Chapter I. Introduction

## I.1 Background

DNA methylation is an important epigenetic process that controls how genes work. It plays a key role in many biological activities, such as embryo development, aging, and cancer. Usually, most DNA methylation is erased during a reprogramming process in early embryos to start a new cycle. However, some areas, like those involved in genomic imprinting, keep their methylation and pass it from parents to children. Also, some studies suggest that environmental factors can cause methylation that escapes this erasure, which might lead to diseases in future generations (Maruyama 2022).

CpG islands (CGIs) are areas in the genome with many CpG sites, often found near gene promoters. Most CGIs stay unmethylated, but some become methylated in specific tissues or during certain processes like imprinting. When a CGI is methylated, it usually stops the nearby gene from being active. In previous studies, researchers tried to predict whether CGI methylation is inherited using DNA sequences. However, many of those methods required histone modification data, which is very difficult and expensive to get from small samples like oocytes or embryos (Maruyama 2022).

To solve this problem, a previous study developed a model called CMIC (CGI Methylation Inheritance Classifier). CMIC uses a Deep Learning approach called Recurrent Neural Networks. Unlike older methods, CMIC can learn features from DNA sequences automatically without needing extra biological data. Because the number of available CGIs was small (only 272), CMIC used a special technique called splitDNA2vec to increase the amount of data and achieved a high accuracy (F-measure of 0.93) (Maruyama 2022).

Although CMIC is very accurate, it could be a black box because it is difficult to understand exactly why the model makes its predictions. The goal of this thesis is to move beyond CMIC by using interpretable machine learning models (Maruyama 2022). By using models that are easier to explain, I aim to find specific biological patterns that control DNA methylation inheritance. This will help us discover new biological insights about how epigenetic information is passed down.In this study, I analyzed 673 genomic regions, which include 396 maintained regions and 277 demethylated regions. For the method, I first converted DNA sequences into numerical data using the k-mer approach (with $k$ ranging from $3$ to $8$). Then, I compared two different models: Random Forest and Logistic Regression. I used 5-fold cross-validation and grid search to find the best settings for these models. Finally, I evaluated the models' accuracy and analyzed which DNA patterns were most important for the predictions.

# Chapter II. Materials and Methods

## II.1 Overview of Analysis

Figure 1 shows the overview of the analysis pipeline in this study. The purpose of this analysis is to predict DNA methylation dynamics (maintenance or demethylation) in early mouse embryos. Specifically, this paper predicts whether specific genome regions maintain methylation or undergo demethylation during the transition from oocyte to blastocyst, using only DNA sequence information. The overall workflow consists of the following four phases:

1. Data Preprocessing: We extract genome regions that show specific methylation dynamics from experimental data and build a training dataset.
2. Feature Extraction: We use the k-mer method to encode DNA sequences into numerical vectors $\mathbf{x} \in \mathbb{R}^{4^k}$.
3. Modeling and Optimization: We select the best algorithm and hyperparameters using grid search with 5-fold cross-validation.
4. Evaluation and Interpretation: We verify the model performance using independent test data. Then, we identify DNA motifs that likely contribute to methylation dynamics by analyzing the coefficients of the trained model.

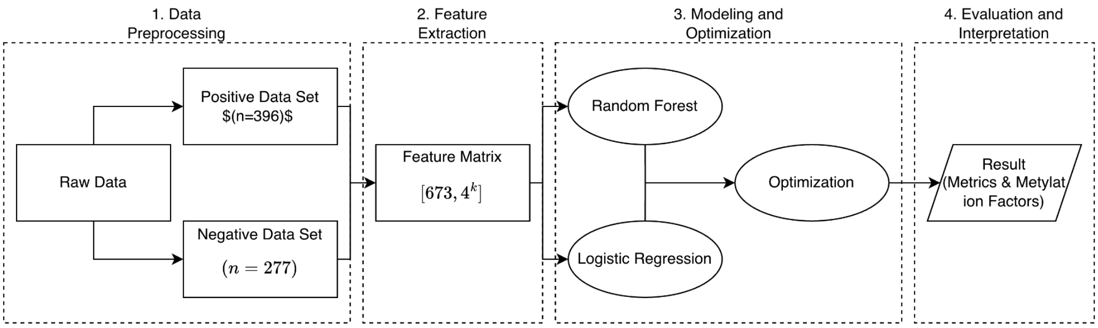


Figure 1: Analysis Pipeline

## II.2 Dataset Source

In this study, we used the data file "for_ML_FGO_PG_202512.xlsx" provided by Assistant Professor Au Yeung. This dataset is based on the records of DNA methylation states in CpG Islands during mouse preimplantation development, which were built by Maruyama et al. (2022).

The Whole Genome Bisulfite Sequencing data, which is the basis of this analysis, comes from two sources. For Fully Grown Oocytes, the data were integrated from several previous studies (Shirane et al., 2013; Maenohara et al., 2017; Au Yeung et al., 2019; Kibe et al., 2021). For blastocysts, the data was an in-house dataset of Au Yeung laboratory.

## II.3 Sample Information

The following information describes the samples and references used in this study.

- Species and Strain: We used the C57BL/6J strain of $Mus\ musculus$. We also used hybrid blastocysts. These blastocysts were obtained via parthenogenetic activation of superovulated Metaphase II (MII) oocytes of C57BL/6J.
- Analysis Stages: We analyzed Fully Grown Oocytes and Blastocysts.
- Genome Information: We used NCBI GRCm38.p6 (mm10) as the mouse reference genome. The definitions of CpG Island regions were based on information from the UCSC Genome Browser database.

## II.4 Data Preprocessing

The data used in this study was preprocessed and organized as a training dataset through the following steps:

1. Read Mapping: Sequence alignment to the mouse reference genome (NCBI GRCm38.p6 / mm10) was conducted using Bismark.
2. Creation of Training Data: Labels were defined based on the changes in DNA methylation from the oocyte stage to the blastocyst stage. The DNA methylation rate $M$ was used as an indicator. First, a common condition for all target regions was set as $M_{\text{oocyte}} \geq 80\%$. Then, the following two groups were defined:
   - Positive Set (Maintenance): $M_{\text{oocyte}} \geq 80\%$ and $M_{\text{blastocyst}} \geq 40\%$ ($n = 396$)
   - Negative Set (Demethylation): $M_{\text{oocyte}} \geq 80\%$ and $M_{\text{blastocyst}} \leq 20\%$ ($n = 277$)

A total of 673 regions were analyzed in this study. Intermediate regions, where the methylation rate in the blastocyst was $20\% < M_{\mathrm{blastocyst}} < 40\%$, were removed from the dataset. DNA methylation in parthenogenetic blastocysts displays a bimodal pattern with peaks at 20% and 40%, which are signs of methylation maintenance and demethylation. This facilitates the efficient extraction of sequence features that determine methylation dynamics within the machine learning model.

## II.5 Feature Extraction (k-mer Analysis)

The k-mer method was adopted to convert DNA sequence information into numerical vectors that machine learning models can process. This method was also used to extract sequence features related to methylation dynamics. In this method, each sequence is divided into continuous substrings of length $k$, and their frequencies are calculated as features. Specifically, the CountVectorizer from the Python scikit-learn library was used for this process. A sliding window process was applied to each DNA sequence to extract substrings of length $k$ by shifting one base at a time. All possible k-mers in the dataset were defined as a vocabulary. By counting the number of occurrences of each k-mer in each sequence, a high-dimensional feature matrix was generated. In this study, $k$ values from $k = 3$ to $k = 8$ were tested to find the optimal length of $k$. The vectorization of k-mers is defined as a process that divides a DNA sequence $S$ into substrings of length $k$ and generates a feature vector $\mathbf{x} = [c_1, c_2, \ldots, c_{4^k}] \in \mathbb{R}^{4^k}$. Here, $c_i$ represents the number of occurrences of the $i$-th k-mer.

## II.6 Machine Learning Algorithms

For the construction of predictive models, the following two types of algorithms with different characteristics were used:

1. Random Forest: This algorithm was selected to verify the predictive performance while considering non-linear interactions between k-mer features.
2. Logistic Regression: This algorithm was adopted to quantitatively evaluate the additive contribution of each k-mer to the maintenance or demethylation of methylation dynamics.

## II.7 Hyperparameter Tuning and Cross-validation

To optimize the generalization performance of the models, a grid search with 5-fold cross-validation was performed. In this study, the parameter search focused on the Logistic Regression model, which showed favorable results in comparative experiments.

The regularization strength $C$ was evaluated in the range of $[0.01, 0.1, 1.0, 10.0]$. This was intended to suppress overfitting and extract motifs that are considered to have a dominant contribution to methylation control. Following common practice in machine learning, a log scale was used for these values. Regarding class_weight, `balanced` or `None` were tested as options. Since there is an imbalance in the number of samples between the Positive group ($n = 396$) and the Negative group ($n = 277$), the `balanced` option was introduced. This was intended to adjust the penalty for misclassifying the minority Negative group and to allow the model to learn the features related to demethylation more accurately. The hyperparameter yielding the highest accuracy is selected.

## II.8 Evaluation Metrics

To evaluate the predictive performance of the models, Accuracy, Precision, Recall, $F_1$-score, and AUPRC (Area Under the Precision-Recall Curve) were calculated.

## II.9 Computational Environment

The analysis was performed in a Python environment (version 3.12.12) on Google Colaboratory. The major libraries used in this study were pandas (version 2.2.2) for data processing, scikit-learn (version 1.6.1) for the implementation of machine learning algorithms, and matplotlib (version 3.10.0) and seaborn (version 0.13.2) for result visualization.

# Chapter III. Results

## III.1 Comparison of Model Accuracy and Motif Trends in Preliminary Experiments

Before full model optimization, a preliminary validation was conducted under the condition of $k = 4$. This condition was chosen to suppress the increase in feature space while capturing the smallest unit of biological motifs. As a result, accuracies of 79.26% for Random Forest and 77.04% for Logistic Regression were obtained. When the feature importance in Random Forest (Table 1) was analyzed, it was confirmed that many of the top k-mers tended to have high frequencies in the Negative group. On the other hand, the sequence catc, which is specific to the Positive group, was also identified.

Table 1: The features Trend of Random Forest at $k = 4$

| k-mer | Importance | Neg(0) Mean | Pos(1) Mean | Tendency |
|---|---|---|---|---|
| cgcc | 0.0497 | 7.2816 | 2.4167 | Demethylation (Neg) |
| cgcg | 0.048 | 5.7798 | 1.8535 | Demethylation (Neg) |
| cccg | 0.0423 | 7.0722 | 2.6717 | Demethylation (Neg) |
| cccc | 0.0374 | 6.7581 | 2.7601 | Demethylation (Neg) |
| ccgc | 0.0369 | 7.9567 | 2.9066 | Demethylation (Neg) |
| cggg | 0.0258 | 6.8303 | 2.8157 | Demethylation (Neg) |
| gccc | 0.0227 | 7.3646 | 3.0581 | Demethylation (Neg) |
| ggcg | 0.0224 | 7.13 | 2.8308 | Demethylation (Neg) |
| gcgg | 0.0182 | 7.7726 | 3.048 | Demethylation (Neg) |
| tccc | 0.0159 | 4.2022 | 1.8788 | Demethylation (Neg) |
| gcgc | 0.015 | 6.9386 | 3.1793 | Demethylation (Neg) |
| gcct | 0.0148 | 3.7798 | 2.6162 | Demethylation (Neg) |
| gggg | 0.0147 | 6.2996 | 2.5909 | Demethylation (Neg) |
| ccgg | 0.0127 | 6.0181 | 2.5101 | Demethylation (Neg) |

| | | | | |
|---|---|---|---|---|
| agcc | 0.0127 | 4.3321 | 2.6439 | Demethylation (Neg) |
| **catc** | **0.0114** | **1.2238** | **1.9545** | **Maintenance (Pos)** |
| gccg | 0.011 | 6.343 | 3.2247 | Demethylation (Neg) |
| ccct | 0.0104 | 3.639 | 1.8258 | Demethylation (Neg) |
| ggcc | 0.0096 | 6.5018 | 3.4924 | Demethylation (Neg) |
| gggc | 0.0087 | 6.8809 | 3.2778 | Demethylation (Neg) |

## III.2 Comparison of k-mer Length and Machine Learning Algorithms

Based on the results of the preliminary experiments, comparison experiments for model accuracy were conducted for $k = 3$ to $k = 8$. The results are shown in Figure 2 below.

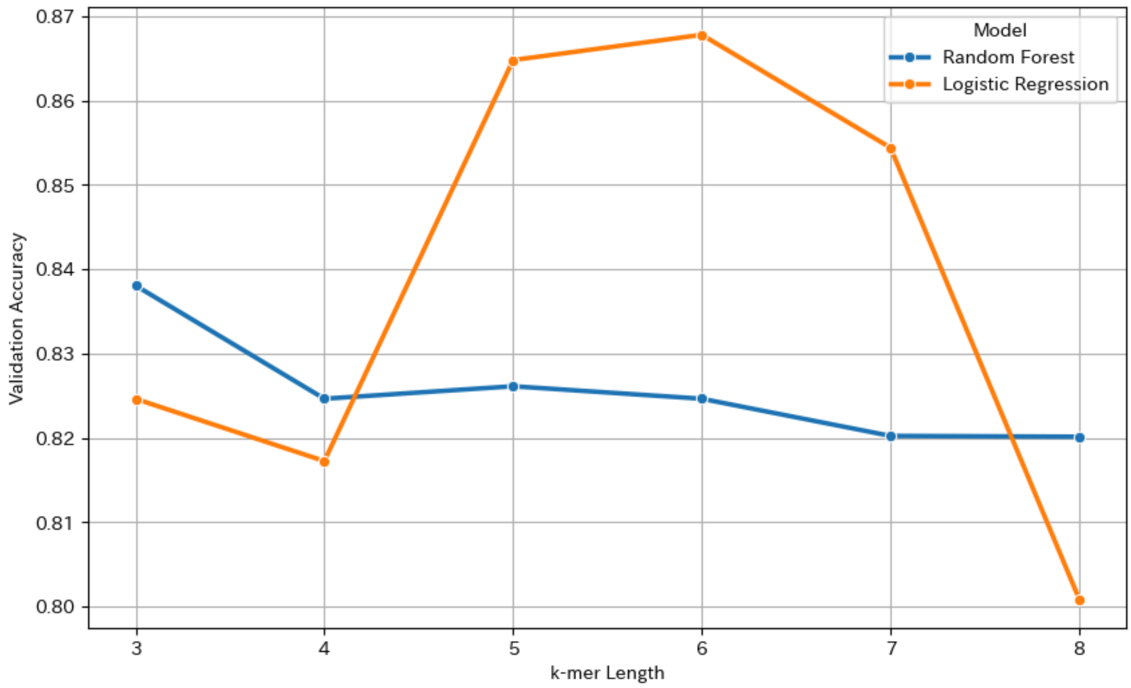


Figure 2

- For $k = 5, 6,$ and $7$, Logistic Regression tended to show higher accuracy than Random Forest.
- For $k = 3, 4,$ and $8$, Random Forest tended to show higher accuracy than

Logistic Regression

- The highest accuracy of 86.78% was recorded at $k = 6$ with Logistic Regression. On the other hand, a decreasing trend in accuracy was observed as $k$ increased to 7 and 8.

Based on the above results, only the Logistic Regression model with k=6 will be used in subsequent experiments.

## III.3 Hyperparameter Tuning and Performance Evaluation of the Optimized Model

Hyperparameter optimization using grid search was performed for the Logistic Regression model with $k = 6$, which showed the highest accuracy in Section 3.2. As a result of the optimization, $C = 0.01$ and class_weight='balanced' were selected as the parameters, with the highest accuracy (83.70%) on the independent test data. The AUC of the ROC (Receiver Operatorating Characteristic) curve for this model was 0.930 (Figure 3). This value suggests that the model has good performance in distinguishing between maintenance regions and demethylation regions.

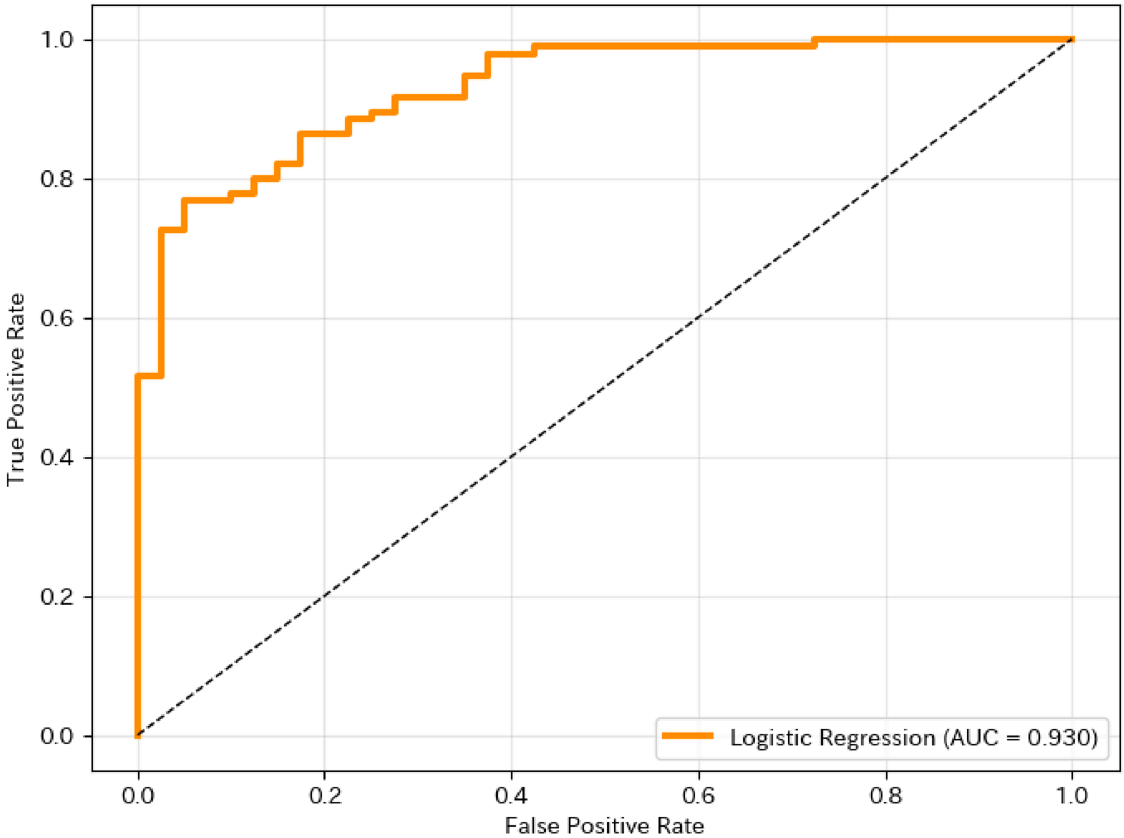


Figure 3: ROC Curve for the Optimal Model

## III.4 Factor Analysis

Through the analysis of the coefficients of the optimized model (Figure 4), the following numerical characteristics were confirmed:

- Negative: Sequences consisting only of C and G, such as ccccgc ($-0.154$) and cccgcc ($-0.153$), recorded high negative scores.
- Positive: Sequences such as ggtcac ($0.077$) and gggtca ($0.060$) were identified with top positive scores. In contrast to the negative side, no sequences consisting only of C and G were found among the features with high scores on the positive side.
- Additionally, the maximum negative coefficient (approximately $-0.15$) was

about twice as large as the maximum positive coefficient (approximately 0.08) in terms of absolute value.

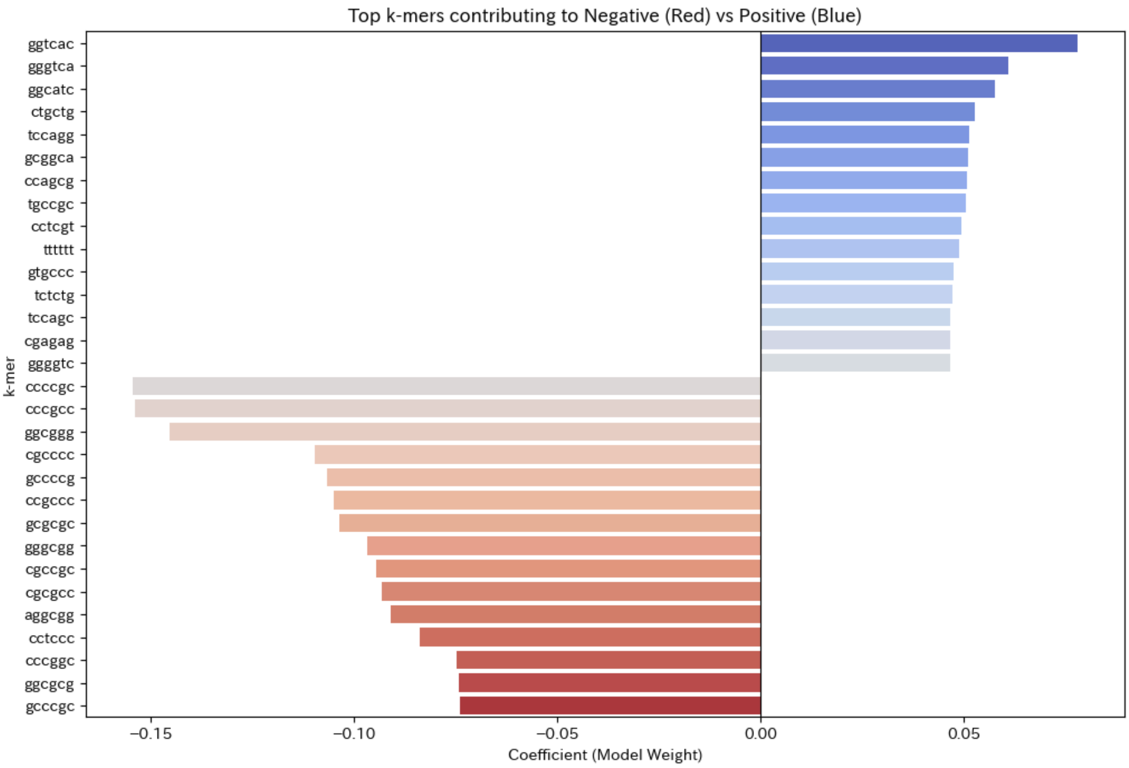


Figure 4: the coefficients of the optimized model

## III.5 Analysis of Misclassifications and Prediction Basis

In this section, the reasons for model predictions and the causes of misclassifications (False Positive; FP, False Negative; FN) are analyzed. This analysis includes identifying overall trends through statistical tests and verifying individual cases using representative sequences.

### III.5.1 Statistical Factor Analysis

For this analysis, the top 15 maintenance and demethylation motifs were selected based on the highest absolute values of their coefficients. The number of these motifs and the CpG counts were then compared among the four groups: TP, TN, FP, and FN. The Mann-Whitney U test, a non-parametric test, was adopted for this comparison (Figure 5). This test was selected because of the small sample sizes ($n_{FP} = 14, n_{FN} = 8$) and the non-normal distribution of the data. A summary of the features for each group is shown in Table 2.

Table 2: A summary of the features

| Category | Count | Maintainance Average(%) | Number of Positive Motif | Number of Negative Motif | CpG Count |
|---|---|---|---|---|---|
| TP | 87 | 80.94931034 | 2 | 1.91954023 | 27.87356322 |
| TN | 26 | 18.46230769 | 2.884615385 | 8.615384615 | 49.26923077 |
| FP | 14 | 63.74357143 | 1.928571429 | 1.928571429 | 1.928571429 |
| FN | 8 | 38.9675 | 2.875 | 2.875 | 2.875 |

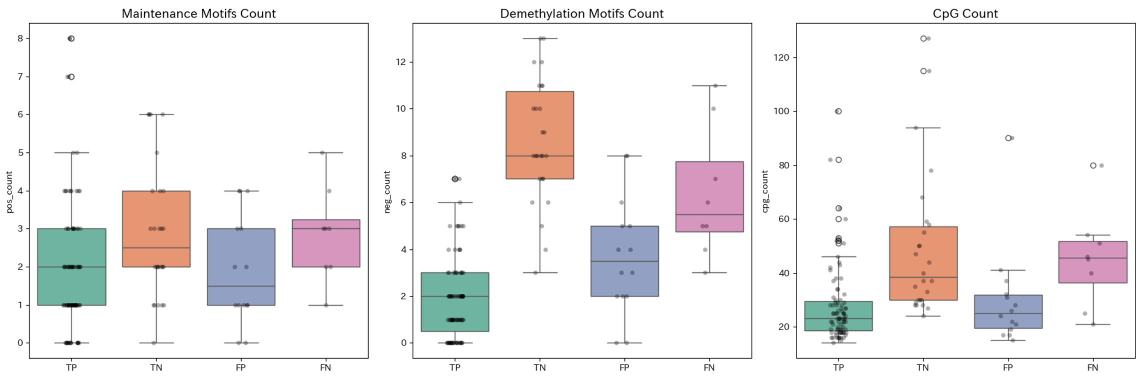


Figure 5: Box plots for the Mann-Whitney U test

According to the results, the TN group showed statistically significant higher values than the TP group in the number of demethylation motifs ($p < 0.001$) and CpG counts ($p < 0.001$). On the other hand, the number of maintenance motifs in the TN group was also significantly higher than in the TP group ($p = 0.013$). This result suggests that DNA methylation dynamics are not determined only by the presence of specific maintenance factors. Instead, the dynamics are likely determined by the balance with demethylation pressure.

In the analysis of misclassified data (FP and FN), the following trends were observed:

1. False Positives (FP): These are cases where demethylation was incorrectly predicted as maintenance. Although these sequences are naturally demethylated, the number of demethylation motifs ($p < 0.001$) and CpG counts ($p = 0.001$) were significantly lower than those in the TN group. Additionally, there was no significant difference when compared to the TP group ($p = 0.749$). This result suggests that the model had difficulty distinguishing these regions from maintenance regions because of a lack of sequence features that promote demethylation.
2. False Negatives (FN): These are cases where maintenance was incorrectly predicted as demethylation. Although these sequences are naturally maintained, the number of demethylation motifs ($p < 0.001$) and CpG counts ($p = 0.004$) were significantly higher than those in the TP group. This indicates that the model tended to predict demethylation when many CpG sites or demethylation motifs were present, even if motifs for maintenance existed.
3. FP and FN often had a maintenance probability closer to 50% than TP and TN, making prediction difficult.

### III.5.2 Case Verification with Representative Sequences

To verify the trends found in the statistical analysis, specific sequences were extracted, and their detailed compositions were checked (Appendix A).

Trends in Correct Data:

- TN: In sequences such as ID 468, there were many demethylation motifs (12 types) and an extremely high CpG count (115). The maintenance probability was calculated as 0.01%, which was a high-confidence prediction.
- TP: In sequences such as ID 54 and 90, maintenance motifs were present, and demethylation motifs were very rare. These sequences were predicted as maintenance with high probability.

Trends in Misclassified Data:

- FN: ID 55 is an example that reflects the statistical trend. This sequence contained 5 types of maintenance motifs. However, it also had 11 types of demethylation motifs and a high CpG count (80). As a result, the maintenance probability was calculated as 49.17%, and the model predicted it as demethylation by a narrow margin.
- FP: In cases like ID 585, no major sequence features for either maintenance or demethylation were found. In such cases, sequences were sometimes predicted as maintenance when the CpG count was relatively low (17). This result may represent one key characteristic of this model: it places greater emphasis on the

presence or absence of negative motifs and CpG counts rather than on the presence or absence of positive motifs. Indeed, in this motif, both positive and negative motifs, along with CpG counts, are low, yet it is still evaluated as positive with a high probability.

# Chapter IV. Discussion

## IV.1 Suggestion of Additivity in Methylation Control

In this study, Logistic Regression (a linear model) showed higher predictive accuracy (86.78% at $k = 6$) than Random Forest, which considers non-linear interactions. This fact suggests that additive effects, where individual motifs contribute independently, are more important for DNA methylation maintenance in early embryos than complex combinations of motifs.

## IV.2 Dominant Influence of Demethylation Pressure

Feature analysis revealed that the peak absolute value of negative coefficients ( $-0.15$) for demethylation was roughly double the maximum value of positive coefficients ($0.08$) for maintenance. This numerical difference suggests that factors for demethylation function more dominantly than factors for maintenance during the global demethylation process.

In this model, `class_weight='balanced'` was used to correct the bias of the imbalanced dataset (Positive: $n = 396$, Negative: $n = 277$). Even after this correction, the negative coefficients remained larger than the positive ones. This supports the strength of the demethylation signal.

Furthermore, several high-ranking demethylation motifs consisted exclusively of cytosine and guanine. This may suggests that the absolute count of CpG sites, rather than the presence of specific protein-binding motifs, serves as a primary determinant for demethylation.

In Section 3.5, some sequences were predicted as demethylated even if they had maintenance motifs. This shows that the protection by maintenance factors can be cancelled out by strong demethylation signals from high CpG density.

## IV.3 Biological Validity of Identified Motifs: Consistency with ZFP57

In the Logistic Regression model, gcggca (6th) and tgccgc (8th) were identified as top k-mers for methylation maintenance. Previous studies show that ZFP57 is an essential protein for methylation maintenance, and its binding motif is tgccgc (Li 2008). Since gcggca is the complementary sequence of tgccgc, the model captures the essential motif regardless of the sequence direction. The fact that the model identified the known ZFP57 binding site without any biological knowledge supports the validity of this analysis.

## IV.4 Validity of k-mer Length ($k = 6$) as a Control Unit

The highest accuracy was recorded at $k = 6$. This matches the length of the core motifs (6bp) recognized by proteins like ZFP57. This suggests the model correctly captured the physical unit of recognition. Statistically, $k = 6$ (4,096 dimensions) is high-dimensional for the sample size ($N = 673$). However, it is estimated that important features were extracted while removing noise because of the strong regularization ($C = 0.01$). Accuracy decreased for $k = 7$ and $8$ likely because data sparsity and overfitting became stronger than the effect of regularization.

## IV.5 Limitations and Future Prospects

While the present model achieved over 80% accuracy in predicting methylation dynamics across varying $k$-mer lengths ($k=3$ to $8$), several limitations remain to be addressed.First, although our findings emphasize the additivity of specific motifs, the high performance of Random Forest across all $k$ values suggests that non-linear interactions between motifs may also contribute to the prediction. If the system were purely additive, Logistic Regression might have yielded superior or near-perfect results. This discrepancy indicates that while the cumulative effect of motifs is a primary driver, complex inter-dependencies between sequences cannot be entirely ignored.

Second, the current analysis primarily focused on the absolute count of motifs and CpG sites, without accounting for their spatial distribution. Factors such as CpG density, the relative distance between motifs, or their specific positioning within the sequence (e.g., proximity to the center versus the distal ends) may significantly influence methylation outcomes. Incorporating these spatial and structural contexts could further refine the model's precision.

Third, while this study focused on simple, highly interpretable models to facilitate motif discovery, there is a clear trade-off between interpretability and predictive power. Future research should explore interpretable deep learning architectures, such as Convolutional Neural Networks with attention mechanisms or saliency maps. Such models could potentially enhance accuracy while still allowing for the identification of functionally relevant sequence features.

Finally, the motifs identified in this study, such as ggtcac for maintenance and ccccgc for demethylation, require rigorous experimental validation. Experimental evidence is essential to confirm whether the identified sequence features are causative drivers of methylation dynamics rather than merely correlative markers.

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# Appendix

## Appendix A

| 配列ID | クラス (正解) | 維持確率 (%) | 維持モチーフ数 | 脱メチル化モチーフ数 | CpGカウント |
|---|---|---|---|---|---|
| 54 | TP | 99.56 | 1 | 0 | 30 |
| 148 | TP | 98.74 | 0 | 0 | 29 |
| 355 | TP | 97.98 | 2 | 1 | 52 |
| 90 | TP | 96.4 | 5 | 0 | 37 |
| 2 | TP | 96.12 | 0 | 0 | 52 |
| 81 | TP | 96.08 | 8 | 5 | 53 |
| 311 | TP | 95.9 | 1 | 0 | 21 |
| 132 | TP | 95.4 | 0 | 0 | 18 |
| 18 | TP | 94.49 | 7 | 4 | 100 |
| 256 | TP | 92.48 | 1 | 1 | 23 |
| 176 | TP | 92.14 | 1 | 2 | 22 |
| 221 | TP | 90.16 | 1 | 0 | 18 |
| 63 | TP | 89.89 | 1 | 0 | 23 |
| 340 | TP | 89.78 | 2 | 2 | 16 |
| 204 | TP | 89.65 | 1 | 1 | 15 |
| 108 | TP | 89.07 | 3 | 4 | 43 |
| 304 | TP | 89 | 5 | 0 | 18 |
| 326 | TP | 88.87 | 4 | 3 | 18 |
| 118 | TP | 88.4 | 3 | 1 | 27 |
| 220 | TP | 88.36 | 1 | 1 | 25 |
| 164 | TP | 87.67 | 0 | 2 | 28 |
| 320 | TP | 87.6 | 1 | 0 | 19 |
| 110 | TP | 87.56 | 2 | 1 | 23 |
| 353 | TP | 87.47 | 1 | 0 | 16 |
| 265 | TP | 86.77 | 1 | 0 | 14 |

| | | | | | |
|---|---|---|---|---|---|
| 257 | TP | 86.25 | 0 | 0 | 18 |
| 331 | TP | 86.1 | 1 | 1 | 29 |
| 133 | TP | 85.79 | 4 | 2 | 25 |
| 76 | TP | 85.52 | 3 | 3 | 34 |
| 318 | TP | 85.34 | 4 | 5 | 38 |
| 390 | TP | 85.07 | 2 | 2 | 18 |
| 327 | TP | 85.03 | 2 | 3 | 31 |
| 101 | TP | 84.86 | 2 | 0 | 41 |
| 135 | TP | 84.81 | 3 | 2 | 20 |
| 78 | TP | 84.7 | 1 | 3 | 16 |
| 10 | TP | 84.58 | 8 | 2 | 82 |
| 271 | TP | 84.53 | 0 | 2 | 16 |
| 296 | TP | 84.29 | 2 | 2 | 27 |
| 248 | TP | 84.13 | 1 | 2 | 22 |
| 212 | TP | 83.94 | 1 | 0 | 22 |
| 41 | TP | 83.5 | 2 | 3 | 24 |
| 281 | TP | 83.03 | 4 | 1 | 25 |
| 363 | TP | 82.98 | 1 | 1 | 18 |
| 335 | TP | 82.97 | 2 | 5 | 38 |
| 367 | TP | 82.93 | 2 | 4 | 25 |
| 86 | TP | 82.81 | 4 | 7 | 44 |
| 274 | TP | 81.96 | 1 | 1 | 17 |
| 165 | TP | 81.92 | 3 | 0 | 18 |
| 31 | TP | 81.73 | 1 | 0 | 22 |
| 145 | TP | 81.47 | 0 | 0 | 17 |
| 49 | TP | 81.36 | 0 | 0 | 16 |
| 82 | TP | 81.25 | 1 | 1 | 26 |
| 56 | TP | 81.12 | 2 | 1 | 23 |
| 69 | TP | 80.65 | 1 | 2 | 18 |
| 51 | TP | 80.43 | 1 | 2 | 26 |

| | | | | | |
|---|---|---|---|---|---|
| 72 | TP | 80.32 | 1 | 2 | 16 |
| 192 | TP | 79.61 | 4 | 5 | 64 |
| 244 | TP | 79.47 | 1 | 1 | 18 |
| 60 | TP | 79.33 | 2 | 1 | 29 |
| 174 | TP | 79.18 | 0 | 2 | 19 |
| 388 | TP | 78.31 | 2 | 1 | 23 |
| 182 | TP | 78.26 | 1 | 2 | 19 |
| 199 | TP | 78.12 | 0 | 0 | 19 |
| 306 | TP | 77.29 | 3 | 4 | 27 |
| 349 | TP | 77.05 | 3 | 3 | 22 |
| 77 | TP | 76.22 | 2 | 2 | 23 |
| 294 | TP | 75.44 | 2 | 3 | 32 |
| 30 | TP | 75.01 | 3 | 2 | 34 |
| 284 | TP | 73.45 | 0 | 1 | 28 |
| 215 | TP | 71.5 | 2 | 7 | 46 |
| 120 | TP | 70.82 | 1 | 0 | 19 |
| 299 | TP | 70.51 | 0 | 1 | 21 |
| 109 | TP | 70.37 | 4 | 3 | 25 |
| 65 | TP | 68.97 | 3 | 6 | 42 |
| 155 | TP | 65.01 | 1 | 5 | 30 |
| 163 | TP | 64.79 | 3 | 4 | 22 |
| 332 | TP | 63.76 | 0 | 2 | 16 |
| 24 | TP | 63.68 | 3 | 5 | 25 |
| 375 | TP | 63.53 | 2 | 0 | 19 |
| 131 | TP | 62.79 | 0 | 2 | 20 |
| 213 | TP | 62.59 | 1 | 2 | 25 |
| 211 | TP | 62.4 | 2 | 3 | 28 |
| 362 | TP | 61.26 | 2 | 1 | 16 |
| 39 | TP | 60.91 | 4 | 7 | 51 |
| 360 | TP | 59 | 4 | 1 | 23 |

| | | | | | |
|---|---|---|---|---|---|
| 44 | TP | 56.93 | 7 | 3 | 60 |
| 181 | TP | 56.12 | 0 | 1 | 28 |
| 468 | TN | 0.01 | 6 | 12 | 115 |
| 523 | TN | 0.19 | 4 | 11 | 78 |
| 607 | TN | 0.96 | 3 | 12 | 68 |
| 612 | TN | 1.6 | 6 | 13 | 127 |
| 453 | TN | 2.26 | 3 | 10 | 47 |
| 488 | TN | 2.42 | 6 | 11 | 59 |
| 462 | TN | 2.49 | 3 | 13 | 37 |
| 640 | TN | 8.32 | 1 | 10 | 33 |
| 451 | TN | 12.62 | 5 | 8 | 28 |
| 603 | TN | 14.2 | 2 | 7 | 27 |
| 671 | TN | 16.2 | 2 | 9 | 30 |
| 620 | TN | 21.8 | 2 | 8 | 50 |
| 500 | TN | 21.93 | 3 | 10 | 55 |
| 479 | TN | 23.64 | 2 | 11 | 37 |
| 631 | TN | 23.72 | 1 | 6 | 58 |
| 541 | TN | 24.83 | 1 | 8 | 40 |
| 573 | TN | 24.96 | 2 | 9 | 24 |
| 642 | TN | 25.39 | 4 | 7 | 94 |
| 557 | TN | 25.9 | 0 | 6 | 44 |
| 527 | TN | 25.97 | 1 | 8 | 28 |
| 602 | TN | 28.42 | 3 | 4 | 30 |
| 434 | TN | 29.2 | 4 | 5 | 50 |
| 669 | TN | 29.77 | 2 | 3 | 28 |
| 457 | TN | 33.23 | 1 | 7 | 29 |
| 630 | TN | 36.34 | 2 | 8 | 30 |
| 660 | TN | 43.65 | 6 | 8 | 35 |
| **585** | FP | 81.59 | 0 | 0 | 17 |
| **625** | FP | 71.77 | 3 | 8 | **90** |

| | | | | | |
|---|---|---|---|---|---|
| **421** | FP | 69.26 | 3 | 3 | 21 |
| **483** | FP | 69.21 | 1 | 2 | 17 |
| **659** | FP | 68.78 | 0 | 0 | 15 |
| **668** | FP | 64.22 | 2 | 4 | 41 |
| **581** | FP | 63.61 | 4 | 2 | 19 |
| **404** | FP | 62.36 | 1 | 5 | 37 |
| **429** | FP | 61.28 | 4 | 4 | 31 |
| **556** | FP | 59.84 | 1 | 3 | 22 |
| **543** | FP | 57.33 | 2 | 5 | 28 |
| **522** | FP | 55.63 | 1 | 2 | 26 |
| **482** | FP | 54.51 | 4 | **8** | 24 |
| **626** | FP | 53.02 | 1 | 6 | 32 |
| **70** | FN | 18.21 | 3 | 7 | 40 |
| **395** | FN | 30.83 | 2 | 6 | 46 |
| **6** | FN | 37.21 | 3 | 5 | 25 |
| **235** | FN | 39.7 | 2 | 3 | 21 |
| **136** | FN | 41.62 | 1 | 10 | 51 |
| **158** | FN | 46.26 | 3 | 4 | 54 |
| **247** | FN | 48.74 | 4 | 5 | 45 |
| **55** | FN | 49.17 | 5 | 11 | 80 |

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## 和文抄訳

DNAメチル化は遺伝子制御において重要な役割を果たすが、その遺伝の要因は完全には解明されていない。深層学習モデルのCMICは高い予測精度を達成したが、そのブラックボックス的な性質が生物学的な理解を妨げている（Maruyama 2022）。本研究では、解釈可能な機械学習モデルであるRandom forestとロジスティック回帰を用いることで、メチル化の継承を制御するDNAモチーフの特定を目的とした。

マウスの673個のゲノム領域（維持群396個、脱メチル化群277個）を対象に、k-mer法（$k = 3$から$8$）を用いて解析を行った。その結果、$k = 6$のロジスティック回帰モデルにおいて、正解率86.78%、AUC 0.930という結果が得られた。

本解析から導かれた結論として、第一に、線形モデルが有効であったことはメチル化継承のプロセスにおいてDNAモチーフの加算的な効果が支配的であることを示唆している。第二に、因子分析の結果、CpG数や特定の脱メチル化モチーフに起因する圧力が、維持シグナルよりも支配的な影響を及ぼしていることが明らかになった。以上の知見は、解釈可能なモデルが、メチル化の遺伝を司る重要な生物学的メカニズムを解明する上で有効であることを示している。